

25.2 Cosmology

Question Paper

Course	CIEA Level Physics
Section	25. Astronomy & Cosmology
Topic	25.2 Cosmology
Difficulty	Medium

Time allowed: 50

Score: /35

Percentage: /100

Question 1a

(a)

(i)

State and explain Hubble's law

[2]

(ii)

State the significance of the term H_0 .

[1]

[3 marks]**Question 1b**

Explain how cosmologists use observations of emission spectra from stars in distant galaxies to determine that the Universe is expanding.

[2 marks]**Question 1c**

Explain how Hubble's law and the idea of the expanding Universe lead to the Big Bang theory of the origin of the Universe.

[3 marks]

Question 1d

Measurements of type Ia supernovae are commonly used to determine a value for the Hubble constant.

The distance from Earth is known for many type Ia supernovae.

(d)

Describe how these distances are used, with other data, to determine the Hubble constant.

[4 marks]

Question 2a

Fig. 1.1 shows the emission spectrum of some wavelengths of light that were observed using a source in a laboratory.



Fig. 1.1

The same wavelengths of light were observed on an emission spectrum from a distant galaxy but were found to be redshifted.

(a)

Describe how you would expect the emission spectrum of the distant galaxy to look compared to the spectrum in Fig. 1.1.

[2 marks]

Question 2b

An astronomer observes that light from a galaxy has been shifted towards the blue end of an emission spectrum.

(b)

State and explain what can be deduced about the motion of the galaxy from this observation.

[2 marks]

Question 2c

Hydrogen is an element of particular interest to cosmologists, as it can emit visible light waves.

A lot of useful information can be obtained by measuring the wavelength of the emitted light from the hydrogen in distant galaxies and comparing them to a laboratory sample. Some sample data is shown in Table 1.2.

Table 1.2

	Wavelength in nm
Laboratory sample	486.14
Galaxy 1a	486.40
Galaxy 1b	484.85

(c)

Galaxy 1a and galaxy 1b are both moving relative to Earth.

Compare the motions of galaxy 1a and galaxy 1b relative to Earth.

[3 marks]

Question 2d

Use data from Table 1.2 to calculate the speed of galaxy 1b relative to Earth.

[3 marks]

Question 3a

An astronomer analyses the light from a distant galaxy.

One of the observed spectral lines of hydrogen has a wavelength of 722 nm. The same spectral line has a wavelength of 656 nm when measured in the laboratory.

(a)

Calculate the factor by which the Universe has expanded since the light was emitted from the source.

[1 mark]

Question 3b

Explain how comparing the values of cosmological redshift for distant galaxies provides evidence for the Big Bang theory.

[4 marks]

Question 3c

Cosmologists commonly quote the recessional velocity v of astronomical objects in relation to the speed of light c . This ratio is known as redshift z :

$$z = \frac{v}{c}$$

The Andromeda galaxy is the closest galaxy to our own Milky Way. Andromeda has a redshift value of $z = -0.001$.

(c)

State the significance of the minus sign and discuss its implication for the Milky Way and Andromeda in the distant future.

[2 marks]

Question 3d

Table 1.1 contains information about two galaxies.

Table 1.1

Galaxy	Red-shift	Recessional velocity / km s^{-1}	Distance / km
NGC 247	8.0×10^{-4}		1.1×10^{20}
NGC 1357		1200	

(d)

(i)

Complete the missing information in Table 1.1.

[2]

(ii)

Use the data to estimate the age, in years, of the Universe.

$$1 \text{ year} = 3.15 \times 10^7 \text{ s}$$

[4]

[6 marks]



Model Answers: Medium

1a

(a)

(i) Hubble's Law is:

- Speed is (directly) proportional to distance **OR** $v = H_0 d$; [1 mark]
- Where speed v is the speed of recession of the galaxy (from the observer) and distance d is the distance of the galaxy away from the observer; [1 mark]

(ii) The significance of the term H_0 is:

- The age of the universe = $\frac{1}{H_0}$; [1 mark]

[Total: 3 marks]

1b

(b) Cosmologists use observations of emission spectra from stars in distant galaxies to determine that the Universe is expanding by:

- If the wavelengths (of spectral lines) are greater (than their measured values on Earth); [1 mark]
- This redshift shows that the stars (in distant galaxies) are moving away from Earth; [1 mark]

[Total: 2 marks]

1c

(c) These ideas lead to the Big Bang theory of the origin of the Universe because:

- (All) parts of the Universe are moving away from each other; [1 mark]
- More distant objects are moving away faster; [1 mark]
- (Therefore) matter must have been close together / very dense in the past; [1 mark]

[Total: 3 marks]

1d

(d) Distance of type Ia supernovae and other data can be used to find the Hubble constant as:

The other data needed is:

- (Recessional) velocity / redshift; [1 mark]
- (Obtained) From measurements of the wavelength of spectral lines (from type Ia supernovae); [1 mark]

To determine a value for the Hubble constant:

- Plot a graph of velocity (on the y-axis) against distance (on the x-axis); [1 mark]
- The Hubble constant H_0 is the gradient of the graph; [1 mark]

[Total: 4 marks]

2a

(a) The redshifted emission spectrum would show:

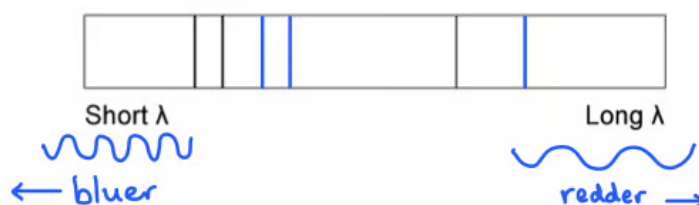
- All the lines moved / shifted towards the 'long λ ' side; [1 mark]
- By the same amount; [1 mark]

[Total: 2 marks]

When an object is moving away from an observer the wavelength of the light increases or becomes longer

As the wavelength of light increases, it moves towards the red end of the spectrum, hence the name red shift

Hence, all the lines will be shifted by the same amount



2b

(b) Blueshift means that:

- The galaxy is moving towards the observer; [1 mark]

This is because:

- Blueshift is a decrease in the observed wavelength of light (emitted from incoming stars and galaxies)

OR

- The light from distant galaxies has a shorter wavelength than the same light observed on Earth; [1 mark]

[Total: 2 marks]

The change in the wavelength of the observed light can tell us about the motion of the galaxy relative to the Earth

- If it is moving **away** from the Earth, the wave will spread out, hence the **wavelength will increase** and appear more **red** (hence the term 'red shift')
- If it is moving **towards** the Earth, the wave will bunch together, hence the **wavelength will decrease** and appear more **blue** (hence the term 'blue shift')

2c

(c) Compare the motions of galaxy 1a and galaxy 1b relative to Earth:

Determine the change in wavelength for galaxy 1a:

- $\Delta\lambda = \lambda - \lambda_0$
- $\Delta\lambda_{1a} = 486.40 - 486.14 = 0.26 \text{ nm}$
- (Therefore,) galaxy 1a is moving away from Earth; [1 mark]

Determine the change in wavelength for galaxy 1b:

- $\Delta\lambda_{1b} = 484.85 - 486.14 = -1.29 \text{ nm}$
- (Therefore,) galaxy 1b is moving towards Earth; [1 mark]

State their relative speeds:

- (And) galaxy 1b is moving faster than galaxy 1a; [1 mark]

[Total: 3 marks]

If $\Delta\lambda$ is **positive** then the galaxy is **moving away** from the Earth. This is also known as **redshift**.

If $\Delta\lambda$ is **negative** then the galaxy is **moving towards** the Earth. This is also known as **blueshift**.

2d

(d) Calculate the speed of galaxy 1b relative to the Earth:

List the known quantities:

- Wavelength in laboratory, $\lambda_0 = 486.14 \text{ nm}$
- Wavelength from galaxy 1b, $\lambda = 484.85 \text{ nm}$
- Speed of light, $c = 3.0 \times 10^8 \text{ m s}^{-1}$
- Change in wavelength for galaxy 1b, $\Delta\lambda = (-)1.29 \text{ nm}$ (from **(c)**)

Rearrange the equation and substitute values, to calculate v :

- $\frac{\Delta\lambda}{\lambda_0} = \frac{v}{c} \Rightarrow v = c \times \frac{\Delta\lambda}{\lambda_0}$ [1 mark]
- $v = (3.0 \times 10^8) \times \frac{1.29}{486.14}$ [1 mark]
- Speed of galaxy 1b = 796 067 = $7.96 \times 10^5 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

3a

(a) The Universe has expanded by:

(d) Calculate the speed of galaxy 1b relative to the Earth:

List the known quantities:

- Wavelength in laboratory, $\lambda_0 = 486.14 \text{ nm}$
- Wavelength from galaxy 1b, $\lambda = 484.85 \text{ nm}$
- Speed of light, $c = 3.0 \times 10^8 \text{ m s}^{-1}$
- Change in wavelength for galaxy 1b, $\Delta\lambda = (-)1.29 \text{ nm}$ (from **(c)**)

Rearrange the equation and substitute values, to calculate v :

- $\frac{\Delta\lambda}{\lambda_0} = \frac{v}{c} \Rightarrow v = c \times \frac{\Delta\lambda}{\lambda_0}$ [1 mark]
- $v = (3.0 \times 10^8) \times \frac{1.29}{486.14}$ [1 mark]
- Speed of galaxy 1b = 796 067 = $7.96 \times 10^5 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

3a

(a) The Universe has expanded by:

- Factor = $\frac{722}{656} = 1.1$ or 10% [1 mark]

[Total: 1 mark]

3b

(b) Comparing values of cosmological redshift for distant galaxies provides evidence for the Big Bang theory because:

Any **one** from:

- The more distant the galaxy, the longer the light travels through space and expands / stretches; [1 mark]
- The more distant the galaxy, the greater the redshift / expansion / stretching of wavelength; [1 mark]

- The more distant the galaxy, the faster it recedes / moves away (from an observer); [1 mark]

This provides evidence for the Big Bang theory because:

- (Increasing redshift suggests that) the universe is expanding; [1 mark]
- (Expansion suggests that) at an earlier point in time **OR** at the beginning / origin of time; [1 mark]
- The universe was a single point; [1 mark]

[Total: 4 marks]

3c

(c) The significance of the minus sign is:

- The light from Andromeda is blueshifted **OR** Andromeda is moving towards us / the Milky Way; [1 mark]

The implication of this is:

- Andromeda will hit / collide with the Milky Way (in the distant future); [1 mark]

[Total: 2 marks]

3d

(d)

List the known quantities:

- 1 year = 3.15×10^7 s
- Speed of light, $c = 3.0 \times 10^8$ m s⁻¹

(i) Complete Table 1.1:

Galaxy	Red-shift	Recessional velocity / km s ⁻¹	Distance / km
NGC 247	8.0×10^{-4}	240	1.1×10^{20}
NGC 1357	4.0×10^{-3}	1200	5.5×10^{20}

- Redshift: $z = \frac{v}{c} \Rightarrow v = cz$
- Recessional velocity of NGC 247: $v = (3.0 \times 10^8) \times (8.0 \times 10^{-4}) = 240\,000 = 240 \text{ km s}^{-1}$ [1 mark]
- Redshift of NGC 1357: $z = \frac{1200 \times 10^3}{3 \times 10^8} = 4.0 \times 10^{-3}$ [1 mark]

Use Hubble's law to calculate the distance of NGC 247:

- Hubble's law: $v = H_0 d$ [1 mark]
- Distance of NGC 247: $H_0 = \frac{v_{247}}{d_{247}} = \frac{v_{1357}}{d_{1357}} \Rightarrow d_{1357} = d_{247} \times \frac{v_{1357}}{v_{247}}$
- Distance of NGC 247: $d_{1357} = (1.1 \times 10^{20}) \times \frac{1200}{240}$
- Distance of NGC 247: $d_{1357} = 5.5 \times 10^{20} \text{ km}$ [1 mark]

(ii) Estimate the age, in years, of the Universe:

- Hubble constant: $H_0 = \frac{v}{d} = \frac{240 \times 10^3}{1.1 \times 10^{20}}$
- Hubble constant: $H_0 = 2.18 \times 10^{-18} \text{ s}^{-1}$ [1 mark]
- Age of Universe: $t = \frac{1}{H_0} = \frac{1}{2.18 \times 10^{-18}} = 4.58 \times 10^{17} \text{ s}$
- Age of Universe: $t = \frac{4.58 \times 10^{17}}{3.15 \times 10^7} = 1.46 \times 10^{10} \text{ years}$ [1 mark]

[Total: 6 marks]

